

Challenge (Habanero)

- Q1.** In the number 4321, the value of the digit 4 is in the place.
(A) ones
(B) tens
(C) hundreds
(D) thousands
(E) ten thousands
- Q2.** What is the standard form of the number $2 \cdot 10^3 + 5 \cdot 10^1 + 9 \cdot 10^0$?
(A) 2000
(B) 2500
(C) 2590
(D) 2599
(E) 3000
- Q3.** A gamer scored 427 points in a level. Rounded to the nearest hundred, the score is _____.
(A) 400
(B) 500
(C) 427
(D) 430
(E) 450
- Q4.** The expanded form of 9421 is _____.
(A) $9 \cdot 10^3 + 4 \cdot 10^2 + 2 \cdot 10^1 + 1 \cdot 10^0$
(B) $9 \cdot 10^3 + 4 \cdot 10^2 + 2 \cdot 10^1 + 1 \cdot 10^1$
(C) $9 \cdot 10^3 + 4 \cdot 10^2 + 2 \cdot 10^0 + 1 \cdot 10^0$
(D) $9 \cdot 10^4 + 4 \cdot 10^3 + 2 \cdot 10^2 + 1 \cdot 10^1$
(E) $9 \cdot 10^4 + 4 \cdot 10^3 + 2 \cdot 10^1 + 1 \cdot 10^0$
- Q5.** In coding, a variable is assigned the value $x = 1032$. The digit 3 in this number has a place value of _____.
(A) hundreds
(B) tens
(C) ones
(D) thousands
(E) ten thousands
- Q6.** A mobile game uses 256 MB of data to download. Rounded to the nearest ten, this is _____ MB.
(A) 250
(B) 260
(C) 256
(D) 270
(E) 300
- Q7.** What is the value of the digit 7 in the number 279,412?
(A) seven thousands
(B) seven hundreds
(C) seven tens
(D) seven ones
(E) seven ten thousands
- Q8.** A digital scoreboard shows a score of 9452. The place value of the digit 9 is _____.
(A) ones
(B) tens
(C) hundreds
(D) thousands
(E) ten thousands

Open Ended Questions (FRQ)

- Q9.** Write the number 6543 in expanded form and standard form. Explain your steps.
- Q10.** Round the number 2198 to the nearest hundred and ten. Show your calculations and explain the reasoning behind each rounding.

Chef's Tips

- Ensure students understand the concept of place value in multi-digit numbers.
- Encourage students to apply the concept to real-world scenarios like coding and gaming.
- Remind students to use the correct mathematical notation when writing numbers in expanded and standard form.